

Project Design Phase-II Technology Stack (Architecture & Stack)

Team ID	SWTID-2026-5404
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: High Impact Incident controls using UI Policy and Client Scripts on ServiceNow

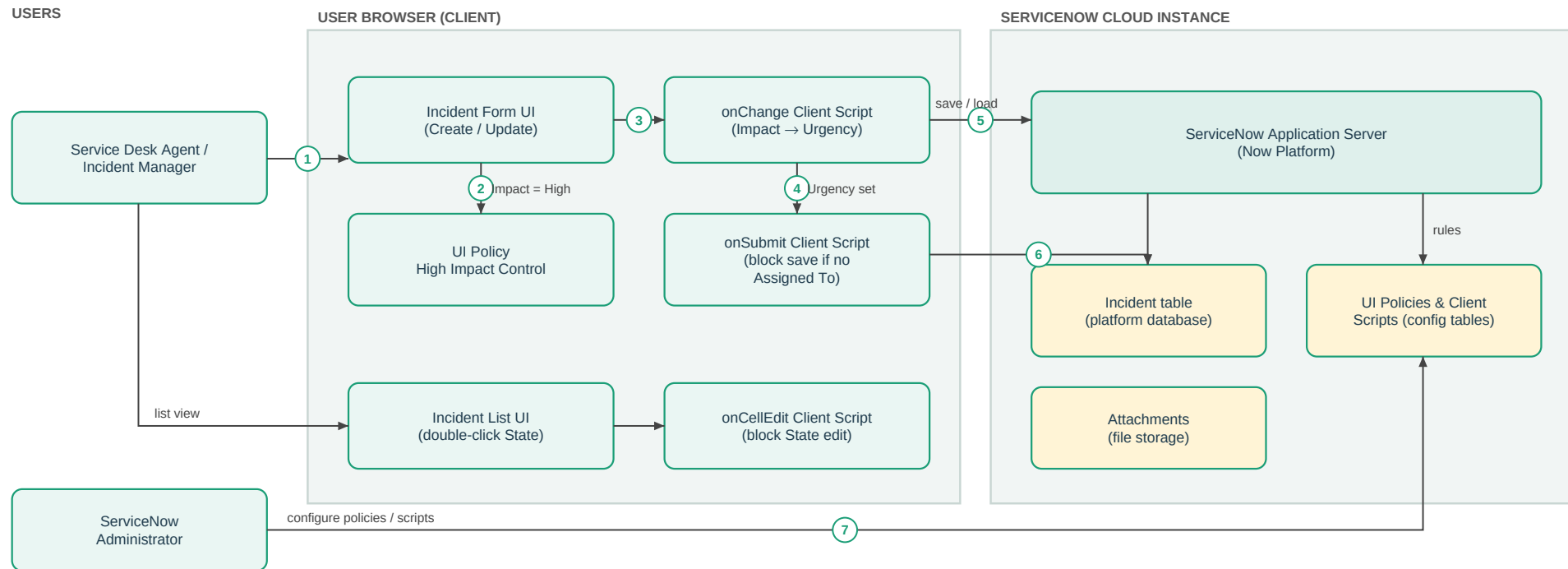


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Incident form and Incident list in the ServiceNow web UI used by agents and managers to create, update and list incidents.	HTML, CSS, JavaScript (ServiceNow platform UI)

S.No	Component	Description	Technology
2.	Application Logic-1	UI Policy <i>High Impact Control</i> on Incident: condition Impact is 1 - High; Assignment group mandatory; Urgency read-only; Reverse if false.	ServiceNow UI Policy & UI Policy Actions
3.	Application Logic-2	onChange Client Script on Impact (sets Urgency to High with an info message) and onSubmit Client Script (blocks save when Impact is High and Assigned To is empty).	JavaScript (g_form / GlideForm client API)
4.	Application Logic-3	onCellEdit Client Script on State that blocks direct State edits from the Incident list and shows an alert.	JavaScript (onCellEdit callback)
5.	Database	Incident [incident] table and configuration tables for UI Policies and Client Scripts.	ServiceNow platform database (relational)
6.	Cloud Database	Database service hosted with the ServiceNow cloud instance (personal developer instance).	ServiceNow Now Platform (cloud)
7.	File Storage	Incident attachments and configuration captured in Update Sets.	ServiceNow attachment storage / Update Sets
8.	External API-1	No external API is required; all controls run client-side. Optional integration channel for incident creation.	ServiceNow REST Table API (optional)
9.	External API-2	Not applicable for this project.	N/A
10.	Machine Learning Model	Not used in this project.	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on ServiceNow cloud instance; accessed from desktop browsers. Local Server Configuration: none – desktop browser (Chrome / Edge) client only. Cloud Server Configuration: ServiceNow-hosted instance (Now Platform).	Cloud (ServiceNow instance)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No custom framework added; controls are plain JavaScript on the platform client API, run with Isolate script enabled.	JavaScript (platform client API)
2.	Security Implementations	Configuration limited to admin / configuration roles; role-based access controls on incident data; isolated client scripts (strict mode, no direct DOM access); HTTPS access. Client-side rules should be backed by server-side rules for API / import paths.	ServiceNow roles & ACLs, Isolate script, HTTPS / TLS
3.	Scalable Architecture	Multi-tier platform (browser, application server, database); policies and scripts are scoped to the Incident table and can be extended to more fields or tables.	ServiceNow Now Platform (3-tier)
4.	Availability	Controls run in the browser as part of the hosted platform, so they are available whenever the Incident form and list are available; no extra server needed.	ServiceNow cloud hosting
5.	Performance	Lightweight scripts; isLoading guard avoids work on form load; no server round-trips (no GlideAjax); UI Policy evaluated instantly on change.	Client-side JavaScript, UI Policy on load

References:

<https://c4model.com/>
<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
<https://www.ibm.com/cloud/architecture>
<https://aws.amazon.com/architecture>
<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>